higher education & training

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COMMUNICATION-ELECTRONICS N6

(8080246)

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This question paper consists of 6 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
COMMUNICATION-ELECTRONICS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

A 600 ohm transmission line requires a composite filter consisting of the following sections:

- Input half-section
- Constant k-section
- m-derived section
- Output half-section

The filter should have a cut-off frequency of 1,9 kHz and the infinite attenuation is 200 Hz above cut-off.

Design the composite filter and draw each section separately. Indicate ALL relevant component values for a T-section on the composite network. **[18]**

QUESTION2

2.1 Network 1 has an output impedance of 560 ohm and network 2 has an input impedance of 420 ohm. It was found by calculation that the inserted attenuator caused an insertion loss of 15 dB. The maximum allowable attenuation should not exceed 6 dB.

Design a T-type asymmetrical attenuator to fulfil the given conditions. Sketch the completed attenuator and insert ALL relevant values. (8)

2.2 The following information for a pi-type (π) asymmetrical attenuator network is given:

Attenuation:	10
Output image impedance:	300 ohms
Series resistor:	4 kilohms

Use the available information to determine the input image impedance. (4)

2.3 Draw the circuit diagram that represents a transmission-line RF-equivalent circuit. (2)

2.4 Name THREE elements that determine the characteristic impedance of a transmission line. (3)

[17]

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QUESTION 3

3.1 A certain transmission line has the following primary line constants:

- R_l
- C_l
- L_l

Derive, from first principles, the formula for the characteristic impedance for this transmission line. (6)

3.2 State the purpose of each of the following transmission components:

3.2.1 Sub

3.2.2 Directional coupler

3.2.3 Balun

3.2.4 Slotted line

(4 × 1) (4)

3.3 Briefly explain how to obtain the SWR on a transmission line by using a Smith chart. (6)

3.4 Which transmission-line components will be used in each of the following cases:

3.4.1 Measuring power in a line

3.4.2 Determining the first voltage maximum from the load

(2 × 1) (2)

[18]

QUESTION 4

4.1 The following data is given for an RF-amplifier:

- Gain: 75
- Output load impedance: 150 kilohms
- Equivalent noise resistance: 5,6 kilohms
- Input noise resistance: 2,2 kilohms
- Frequency range: 60 MHz to 90 MHz
- Temperature: 27 °C
- Antenna impedance: 50 ohm
- Boltzmann's constant: $1{,}38 \times 10^{-23}$ J/K

Determine:

4.1.1 The noise voltage generated (3)

4.1.2 The noise figure (4)

4.2 Name TWO sources of extraterrestrial noise. (2)

4.3 The following information concerning a two-stage amplifier is given:

	STAGE 1	STAGE 2
Gain	10	15
Input resistance	9 kΩ	20 kΩ
Equivalent resistance	12 kΩ	15 kΩ
Output resistance	18 kΩ	30 MΩ

Determine:

4.3.1 The equivalent noise resistance (5)

4.3.2 The output noise voltage for a bandwidth of 9 MHz and a temperature of 25 °C (2)

4.4 Explain the equivalent noise resistance of an amplifier or receiver according to noise figure. (2)

[18]

QUESTION 5

5.1 Draw a neat, fully labelled block diagram of a single-channel PCM-system. (10)

5.2 Give THREE advantages and THREE disadvantages of PCM over other methods of communication. (3 + 3) (6)

[16]

QUESTION 6

6.1 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (6.1) in the ANSWER BOOK.

The gain of an antenna effectively increases the effective aperture of the antenna. (1)

6.2 The antenna of a transmitter station has an effective aperture of 12 m^2 and radiates energy at a frequency of 120 MHz.

Calculate:

6.2.1 The power received for a power density of $1,2 \times 10^{-12}$ W/m^2 (2)

6.2.2 The gain of the antenna in dB (4)

6.3 Define each of the following terms referring to antennae:

6.3.1 Directive gain

6.3.2 Receiver antenna

6.3.3 Transmitter antenna

(3 × 2) (6)
[13]

TOTAL: 100

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FORMULA SHEET

1. FILTERS

Low-pass

$$f_c = \frac{1}{\pi \sqrt{L \,.\, C}} \qquad Z_o = \sqrt{\frac{L}{C}}$$

High-pass

$$f_c = \frac{1}{4 \,.\, \pi \sqrt{L \,.\, C}} \qquad Z_o = \sqrt{\frac{L}{C}}$$

Low-pass m-derived sections (also for half-sections):

T-section **π-section**

$$\text{Series } L = \frac{m \,.\, L}{2} \qquad \text{Shunt } C = \frac{m \,.\, C}{2}$$

$$\text{Shunt } C = m \,.\, C \qquad \text{Series } L = m \,.\, L$$

$$\text{Shunt } L = \frac{1 - m^2}{4 \,.\, m} \,.\, L \qquad \text{Series } C = \frac{1 - m^2}{4 \,.\, m} \,.\, C$$

$$m = \sqrt{1 - \frac{f_c^2}{f_\infty^2}} \qquad L = \frac{Z_o}{\pi \,.\, f_c} \qquad C = \frac{1}{\pi \,.\, Z_o \,.\, f_c}$$

High-pass m-derived sections (also for half-sections):

T-section **π-section**

$$\text{Series } C = \frac{2 \,.\, C}{m} \qquad \text{Series } C = \frac{C}{m}$$

$$\text{Shunt } C = \frac{4 \,.\, m}{1 - m^2} \,.\, C \qquad \text{Series } L = \frac{4 \,.\, m}{1 - m^2} \,.\, L$$

$$\text{Shunt } L = \frac{L}{m} \qquad \text{Shunt } L = \frac{2 \,.\, L}{m}$$

$$m = \sqrt{1 - \frac{f_\infty^2}{f_c^2}} \quad L = \frac{Z_o}{4 \,.\, \pi \,.\, f_c} \qquad C = \frac{1}{4 \,.\, \pi \,.\, Z_o \,.\, f_c}$$

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Band-pass sections:

$$\text{Series } L = \frac{Z_o}{2 \,.\, \pi \,[f_2 - f_1]} \qquad \text{Series } C = \frac{f_2 - f_1}{2 \,.\, \pi \,.\, Z_o \,.\, f_1 \,.\, f_2}$$

$$\text{Shunt } L = \frac{Z_o \,[f_2 - f_1]}{4 \,.\, \pi \,.\, f_1 \,.\, f_2} \qquad \text{Shunt } C = \frac{1}{\pi \,.\, Z_o \,[f_2 - f_1]}$$

Band-stop sections:

$$\text{Series } L = \frac{Z_o \,[f_2 - f_1]}{2 \,.\, \pi \,.\, f_1 \,.\, f_2} \qquad \text{Series } C = \frac{1}{2 \,.\, \pi \,.\, Z_o \,[f_2 - f_1]}$$

$$\text{Shunt } L = \frac{Z_o}{4 \,.\, \pi \,.\, [f_2 - f_1]} \qquad \text{Shunt } C = \frac{f_2 - f_1}{\pi \,.\, Z_o \,.\, f_1 \,.\, f_2}$$

2. ATTENUATORS

Symmetrical:

Bridged T-type

$$Z_B = Z_o \,[N - 1]$$

$$Z_c = \frac{Z_o}{N - 1}$$

$$Z_A = Z_o \quad \text{where } dB = 20 \log N$$

Asymmetrical:

L-type

$$Z_A = \sqrt{R_1 \,(R_1 - R_2)}$$

$$Z_B = \sqrt{\frac{(R_1 \,.\, R_2^2)}{R_1 - R_2}}$$

T-type

$$Z_1 = R_1 \left[\frac{N^2 + 1}{N^2 - 1}\right] - 2\sqrt{R_1 \,.\, R_2} \left[\frac{N}{N^2 - 1}\right]$$

$$Z_2 = 2\sqrt{R_1 \,.\, R_2} \left[\frac{N}{N^2 - 1}\right]$$

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$$Z_3 = R_2\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_1 . R_2}\left[\frac{N}{N^2-1}\right] \text{ where } dB = 20 \log N$$

π-type

$$Z_A = R_1\left[\frac{N^2-1}{N^2-2 . N . S+1}\right]$$

$$Z_B = \frac{\sqrt{R_1 . R_2}}{2}\left[\frac{N^2-1}{N}\right]$$

$$Z_c = R_2\left[\frac{N^2-1}{N^2-2 . N/S+1}\right] \quad \text{where } dB = 20 \log N \quad \text{and} \quad S = \sqrt{\frac{R_1}{R_2}}$$

Insertion loss = $20 \log \frac{I_1}{I_2}$

3. TRANSMISSION LINES

$$Z_o = \sqrt{Z_{o/c} . Z_{s/c}} \qquad Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\gamma = \sqrt{[R + j\omega L][G + j\omega C]} \qquad \gamma = \alpha + j\beta$$

$$\lambda = \frac{2 . \pi}{\beta} \quad V = \frac{\omega}{\beta} \qquad t = \frac{d}{v}$$

L.G = C.R

$$VSWR = \frac{Z_R}{Z_o} \text{ or/of } \frac{Z_o}{Z_R}$$

Also $VSWR = \frac{1+\rho}{1-\rho}$

$$\rho = \frac{Z_R - Z_o}{Z_R + Z_o}$$

$$\rho = \frac{V_{(reflected/\ weerkaatste)}}{V_{(incident/\ insident)}}$$

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Normalised impedance

$$Z_o^{'} = \frac{R}{Z_o} \pm j\,\frac{X}{Z_o}$$

4. NOISE

$$E_n = \sqrt{4KT\Delta fR} \qquad E_n = \sqrt{4KT\Delta f\ [R_1 + R_2]}$$

$$R_{eq} = R_A + \frac{R_B}{A_1^2} + \frac{R_c}{A_1^2\ .\ A_2^2} \text{ where } R_A = R_{1in} + R_{1eq}$$

$$R_B = \frac{R_{1(out/uit)}\ .\ R_{2(in)}}{R_{1(out/uit)} + R_{2(in)}} + R_{2(eq)}$$

$$R_c = R_{2out/uit}$$

$$F = 1 + \frac{R_{eq}^{'}}{R_a} \qquad R_{eq}^{'} = R_{eq} - R_t$$

$$F = \frac{S/N_{(in)}}{S/N_{(out/uit)}} \qquad F = 1 + \frac{T_{eq}}{T_o}$$

$$C = \Delta f \log_2 [1 + S/N] \text{ where } dB = 10 \log S/N$$

5. MODULATION

$$m = \frac{V_m}{V_c} \qquad \eta = \frac{P_{dc}}{P_c} \times 100\%$$

$$P_{dc} = I_{mean} \times V_{dc} \qquad P_{\text{mod}} = \frac{P_{SB}}{\eta}$$

$$P_{SB} = \frac{m^2\ .\ P_c}{2}$$

$$P_T = P_c + P_{SB}$$

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6. ANTENNAE

$$P_r = P_d \times A \quad A = \frac{\lambda^2}{4 \,.\, \pi} \qquad \lambda = \frac{c}{f}$$

$$P_r = P_d \times A \,.\, G_r \quad \text{where } dB = 10 \log G$$

$$P_d = \frac{E^2}{120 \,.\, \pi} \qquad P_d = \frac{P_t \,.\, G_t}{4 \,.\, \pi \,.\, d^2}$$

$$P_d = \frac{30 \,.\, G \,.\, P_t}{120 \,.\, \pi \,.\, d^2} \qquad P_t = \frac{[4 \,.\, \pi]^3 \,.\, d^4 \,.\, P_r}{G_t \,.\, G_r \,.\, Q \,.\, \lambda^2}$$

$$P_r = \frac{10 \,.\, P_t \,.\, G_t}{4 \,.\, \pi \,.\, d^2} \qquad E_r = \frac{\sqrt{30 \,.\, P_r}}{d}$$

$$E_S = \frac{\sqrt{30 \,.\, P_t \,.\, G_t}}{d} \qquad A = \frac{G \,.\, \lambda^2}{4 \,.\, \pi}$$

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